Shivraj Bakare

Final Year Undergraduate, IIT Kanpur
Major: Materials Science and Engineering

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Year	Qualification	School/Institution	CPI/%
2023 - Present	B.Tech in MSE	Indian Institute of Technology Kanpur	**7.9**/10
2022	MSBSHSE (XII)	Lalasheth Bihani Vidyamandir Prashala, Rahuri	84.17%
2020	MSBSHSE (X)	Savitribai Phule Madhyamik Vidyalaya, Rahuri	97.0%

Scholastic Achievements

- Achieved a peak rating of **1441 (Specialist)** on **Codeforces**, showing high proficiency in competitive programming *2026*
- Scored **99.72 percentile** in **MHT-CET** among **250k+** participants, conducted by the Maharashtra State CET Cell *2023*
- Secured **State Rank 41** in the **Manthan Talent Search Examination**, conducted by the Maharashtra State Council *2017*

Work Experience

MyConsciOS *Jul'26 - Ongoing*
AI/ML Engineering Intern

- Engineered **LamarX**, a hybrid deterministic and AI decision-intelligence engine (16-state symbolic mapper, 12-house geomantic alignment, Witness/Judge synthesis, 6-factor confidence scoring) via **FastAPI** & **PostgreSQL** with 87 Alembic migrations.
- Built a dual-tier **NLP humanizer** integrating **OpenAI** with a rule-based fallback; fixed silent-disabling & non-contextual response bugs via config refactoring & system-prompt engineering.
- Built a semantic anti-abuse answer-lock using **OpenAI text-embedding-3-small** (1536-dim) and cosine similarity ($\geq$**0.84**) with category-specific windows to prevent impulsive re-querying.
- Integrated a **Beta-Binomial** Bayesian calibration model with **Brier Score** validation and **10-bin calibration** curves for continuously calibrating decision confidence against user feedback.
- Shipped the full-stack platform application & architected **ConsciOS**, a microservices Decision-OS SaaS with API Gateway, Docker, Redis rate limiting & OpenAI Responses **JSON** outputs.

Research Experience

Declarative Sovereign LLM Fine-Tuning *Mar'26 - Aug'26*
Research Intern, under Prof. Philip Treleaven, University College London

- Researched a **privacy-preserving**, **sovereignty-aware** architecture for fine-tuning **LLMs** in regulated financial systems, covering data localisation, security, auditability & full compliance.
- Integrated **DP-SGD**, **Federated Learning**, Federated Computing as Code, Value-Identity Model and **LoRA**/**QLoRA** into a declarative architecture separating governance/privacy planes.
- Engineered an integration pipeline harmonizing **cryptographic primitives** with fine-tuning workflows, keeping **DP-induced latency** within acceptable thresholds for **real-time applications**.
- Compared privacy frameworks across high-frequency-trading & retail-banking scenarios; built a **Governance-as-Code** layer automating compliance auditing, tested against poisoning attacks.
- Co-authored *"A Declarative Architecture for Sovereignty-Aware LLM Fine-Tuning in Regulated Financial Systems"* (**SSRN**), projected to avoid Rs. 50-200 Cr in total custom AI training costs.

Technical Skills

- **Languages:** C, C++, Python, SQL, HTML, CSS, JavaScript
- **Frameworks & Tools:** NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, PyTorch, Next.js, React, Git, GitHub, Excel, AutoCAD

Relevant Courses

Fundamentals of Computing	AI, ML, DL & Applications
Computational Methods in MSE	Materials Informatics
Application of Calculus	Linear Algebra
Quantum Physics	Classical Electrodynamics

Extra-curricular Activities

- **Technical:** Secured **2nd Position** in AIRIDDLE, a design and flight challenge by the Aeromodelling Club at IIT Kanpur.
- **Cultural:**Received **Gold Medal** in **Inter IIT 7.0** Group Dance (overall dance trophy), **Antaragni'24** Jitterbug, **Galaxy'25**.
- **Social:** Completed **NCC Grade A** certification (**Ministry of Defence**) & participated in 10-day Combined Training Camp.

Key Projects

Explainable AI-Generated Image Detection *Jan'26 - Apr'26*
Undergraduate Project, under Prof. A. Behera

- Engineered an explainable AI pipeline for image forensics, performing Real/Fake classification and multi-artifact localization across modern generators (**SD 3.5**, **PixArt-α**, **Adobe Firefly**).
- Combined a **DINOv2** + **RBF-SVM** classifier with a **16-CNN ensemble** (ResNet50/DenseNet121) & **JINA-CLIP-V2** base model to detect **70+** synthetic artifact types in production.
- Developed a **5-tier Priority Fusion Algorithm** and weighted scoring engine to unify multi-model predictions into a ranked artifact list while managing VRAM constraints during inference.
- Integrated a 4-bit quantized **VLM** (Ovis1.6-Gemma2-9B) for zero-shot natural-language explanations, achieving **0.182 IoU** test score across a manually annotated **2,750-image** dataset.

ISTVT DeepFake Detection *May'25 - Jul'25*
EE656 Course Project, under Prof. N. K. Verma, Dept. of EE

- Developed a highly robust, interpretable deepfake detection system using **ISTVT** to capture fine-grained **spatial artifacts** and **temporal inconsistencies** across varied compression levels.
- Built a preprocessing **PyTorch** pipeline using **MTCNN** for aligned face cropping, converting videos into normalized and fixed-length frame sequences to preserve **temporal continuity**.
- Extracted spatial embeddings via pretrained **Xception** backbone & applied a **Tokenizer** with spatial-temporal self-attention blocks & feature subtraction to highlight manipulation artifacts.
- Trained using **BCE** loss, achieving **95.32% accuracy** & **97.24% AUC** on FaceForensics++, visualizing attention maps centered on altered facial regions (eyes, mouth, jawline) at evaluation.

Sanskrit Real-Time ASR Engine *May'25 - July'25*
Summer Project, under Prof. Rajesh M. Hegde, Dept. of EE

- Fine-tuned **Wav2Vec2** for CTC in **PyTorch** on Sanskrit dataset speech audio, processing **16kHz** inputs using **soundfile** and **librosa** to achieve **90%+** transcription accuracy benchmarks.
- Built a high-throughput **FastAPI** server backend service with CORS, integrating **Flan-T5-base** alongside **transcribe** and **feedback** endpoints for real-time human-in-the-loop correction.
- Engineered responsive user interfaces using **Next.js** with **Tailwind CSS** for web audio streaming and a multithreaded **Tkinter** desktop application for microphone and file **transmissions**.

CATGNN for Heat and Crystal Vibrations *May'25 - July'25*
Summer project, under Prof. Somnath Bhowmick, Dept. of MSE

- Predicted lattice thermal conductivity using **ICOHP**/**ICOBI** bonding descriptors with GNNs, generating descriptors for **4,500+** crystals from OQMD via DFT (VASP) & LOBSTER.
- Developed **CATGNN** in **PyTorch** and PyTorch Geometric using spherical harmonics to predict ICOHP/ICOBI from crystal graphs, training on a 75/15/10 split with **AdamW** at lr = 0.001.
- Achieved $\mathbf{R^2}$ of **97.4%** (ICOHP) & **97.8%** (ICOBI), predicting LTC for **200k+ materials** to accelerate thermoelectrics.

Positions of Responsibility

Secretary & Team Member *Apr'24 – Apr'25*
KOS Dance Club, Media & Cultural Council, IIT Kanpur

- Directed a **40-member contingent** and trained **50+ freshers**, serving as core choreographer to secure top inter-college podium finishes and lead public performance campaigns at Z Square Mall.